

Bo-Yu Chen 陳柏昱

Researcher and builder working across music AI, creative tools, musical interfaces, and human-computer interaction.

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RESEARCH INTERESTS

Human-Computer Interaction (HCI) · Music Information Retrieval (MIR) · AI Agents · Creative Support Tools · Musical Interfaces

EDUCATION

National Taiwan University (NTU)
M.S. in Networking and Multimedia (Dept. of CSIE)

- Master's Thesis:** *FlueBricks: A Modular Toolkit for Interactive Flute-like Instrument Systems* [C1] (Advisor: Prof. Lung-Pan Cheng)
- Thesis Honors:** Winner of **TAICHI 2026 Master's Thesis Honorable Mention Award** Top 16.6%

Sep 2021 – Aug 2025
Taipei, Taiwan

National Taiwan University of Science and Technology (NTUST)
B.S. in Electrical Engineering

Sep 2013 – Jun 2018
Taipei, Taiwan

HONORS & AWARDS

Master's Thesis Honorable Mention Award · TAICHI Thesis Award for *FlueBricks* (Top 16.6%, published as [C1])

Best Paper Award 🏆 · Taiwan Computer-Human Interaction Society (TAICHI) for *FlueBricks* [C1]

Overseas Visit Delegate · Epoch Foundation Young Entrepreneurs of the Future (Top 9%, 16/175)

3rd Prize · Ministry of Economic Affairs (MOEA) GetFresh Creativity Competition

2026

2025

2017

2017

RESEARCH EXPERIENCE

Computational Physicality Lab, National Taiwan University
Graduate Research Assistant (Advisor: Prof. Lung-Pan Cheng)

- Designed FlueBricks**, a modular physical-computing toolkit enabling acoustic replication and customizable design of flute-like instruments as Master's thesis research. [C1]
- Engineered parametric CAD assemblies in Fusion 360** for digital fabrication, and conducted user evaluations across musician skill levels to validate expressive playability.

Sep 2021 – Aug 2025
Taipei, Taiwan

Creative AI Lab, Sony Group Corporation
Research Intern (Advisors: Dr. Yuki Mitsufuji, Dr. Marco A. Martínez Ramírez, Wei-Hsiang Liao)

- Developed DJTransGAN**, the first generative model for automated cross-track DJ transitions using DDSP and GANs. [C2]
- Open-sourced the full pipeline**, gathering **120+ GitHub stars** and broad community interest.

Nov 2020 – Sep 2021
Tokyo, Japan (Remote)

Music and AI Lab, Academia Sinica
Research Assistant (Advisor: Prof. Yi-Hsuan Yang)

- Pioneered loop-based music AI research** spanning compatibility assessment, generation, and benchmarking. [C3] [C4] [C5]
- Mentored 5+ junior researchers**, co-authoring 4 papers and establishing reproducible research codebases.

Jul 2019 – Sep 2023
Taipei, Taiwan

WORK EXPERIENCE

Rhythm Culture Corporation 

Technical Lead

- Architecting Rytho**, a local-first creative workspace ("notes with ears") empowering Human-AI co-creation, serving as the central hub coordinating autonomous agent abilities within the **Rytho ecosystem**.
- Led R&D on LLM-driven music agents**, developing conversational feedback agents for music production and reasoning systems for chord analysis. [W1] [W2]
- Managed engineering teams for government R&D projects** (TAICCA, Taipei City) and **TMC-CL1**, a web music interface commissioned by Taipei Music Center.

Jul 2023 – Present
Taipei, Taiwan

- **Architected interactive web audio interfaces and networked performance systems**, developing a vocal sampler for Vocal Asia (Japan) [A2] and a p5.js/Node.js WebSocket engine bridging mobile audiences to Max/MSP stage synthesis [A4].
- **Engineered generative audio-visual systems** in collaboration with electronic music artists, delivering responsive sound-visual synchronization pipelines for MIDEM (France). [A3]

OPEN-SOURCE SOFTWARE & TOOLKITS

OpenMIRLab

Nov 2025 – Present

Founder & Lead Maintainer

Open-Source

- **Packaged reproducible inference suites for 30+ SOTA MIR models** (source separation, transcription, chord estimation, beat tracking), resolving upstream dependency decay across PyTorch and TensorFlow.
- **Standardized unified inference APIs and checkpoint registries**, stripping heavy training dependencies into lightweight, pip-installable packages with verified model outputs.

PUBLICATIONS

C1 FlueBricks: A Construction Kit of Flute-like Instrument for Acoustic Reasoning

Bo-Yu Chen, Chiao-Wei Huang, Lung-Pan Cheng

ACM Conference on Human Factors in Computing Systems (CHI 2026)

Acceptance ~25%

TAICHI 2025 · Best Paper



W1 Enhancing Automatic Chord Recognition through LLM Chain-of-Thought Reasoning

Chih-Cheng Chang, Bo-Yu Chen, Lu-Rong Chen, Li Su

ISMIR 2025 Workshop on Large Language Models for Music (LLM4Music)



W2 AI TrackMate: Finally, Someone Who Will Give Your Music More Than Just "Sounds Great!"

Yi-Lin Jiang, Chia-Ho Hsiung, Yen-Tung Yeh, Lu-Rong Chen, Bo-Yu Chen

Conference on Neural Information Processing Systems (NeurIPS 2024) – Creative AI Track

Acceptance 31%



C2 Automatic DJ Transitions with Differentiable Audio Effects and Generative Adversarial Networks

Bo-Yu Chen, Wei-Han Hsu, Wei-Hsiang Liao, Marco A. Martínez Ramírez, Yuki Mitsufuji, Yi-Hsuan Yang

IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2022)

Acceptance 47%



C3 Exploiting Pre-trained Feature Networks for Generative Adversarial Networks in Audio-Domain Loop Generation

Yen-Tung Yeh, Bo-Yu Chen, Yi-Hsuan Yang

International Society for Music Information Retrieval Conference (ISMIR 2022)

Acceptance 43%



C4 A Benchmarking Initiative for Audio-Domain Music Generation Using the Freesound Loop Dataset

Tun-Min Hung, Bo-Yu Chen, Yen-Tung Yeh, Yi-Hsuan Yang

International Society for Music Information Retrieval Conference (ISMIR 2021)

Acceptance 34%



C5 Neural Loop Combiner: Neural Network Models for Assessing the Compatibility of Loops

Bo-Yu Chen, Jordan Smith, Yi-Hsuan Yang

International Society for Music Information Retrieval Conference (ISMIR 2020)

Acceptance 38%



INTERACTIVE ART & INSTALLATIONS

- A1 Iridescent Vision**   2019
Developed a web-based interactive music video and light installation exploring human civilization and imagined future technology, with DJ Sonia Calico and visual artist Celso Urroz · FINAL Club (Taipei)
- A2 Taptap**  2019
Developed a web-based musical interface that lets visitors play music with a cappella samples, in collaboration with O-KAI SINGERS · Vocal Asia Festival (Kagawa, Japan)
- A3 FusionTaiwanBeat**  2019
Built a web-based interactive audiovisual system showcasing Taiwanese electronic music, with DJ Sonia Calico and visual artist Celso Urroz · MIDEM (Cannes, France)
- A4 Duplex Transmission**  2019
Built a web-based audience-performer interaction system using p5.js, Tone.js, Node.js (WebSockets), and Max/MSP, with new media artist Po-Hao Chi · Songshan Cultural Park (Taipei)
- A5 Mimo No.Tent**  2019
Created a virtual planetarium that brings a starry sky inside a tent, with designers Yi-Yun Liao and Jing-Ting Huang · TaiwanTech Design Grad Show (Taipei, Taiwan)
- A6 YuYu**  2018
Developed an AR service for delivering customized messages through Chinese pastries, with designer Ya-Cin Siao · TaiwanTech Design Grad Show (Taipei, Taiwan)
- A7 Message Container**  2017
Designed a breath-driven immersive installation providing haptic, sound, and visual feedback, with new media artist Wen-Fei Zhang · TaiwanTech Design Grad Show & Venue (Taipei, Taiwan)

ACADEMIC SERVICE & REVIEWING

- International Society for Music Information Retrieval Conference (**ISMIR**) 2020 – 2026
- IEEE International Conference on Acoustics, Speech, and Signal Processing (**ICASSP**) 2025
- Conference on Neural Information Processing Systems (**NeurIPS**) – Creative AI Track 2025

TEACHING EXPERIENCE

- BLND Creative Coding School** Aug 2020 – Jul 2023
Instructor Taipei, Taiwan
- Taught generative AI art (GANs, Stable Diffusion workflows) and physical computing with Arduino to creative practitioners.
- OrangeApple Children Programming School** Jul 2015 – Sep 2019
Instructor Taipei, Taiwan
- Taught algorithms and C++/JS to K-12 students; coached high school students for the APCS examination.

EXTRACURRICULAR ACTIVITIES

- Musical Theater: Rose of the Endless Night**   Feb 2015 – Jun 2015
Director, Music Theater Course Final Public Performance (NTUST) Taipei, Taiwan
- Authored a stage adaptation of a Taiwanese musical and directed 60+ cast and crew members for a live stage production with 400+ audience.
- NTUST Men's Football Team** Jul 2016 – Jul 2017
Vice Captain, Position: Defensive Midfielder / Right-Back Taipei, Taiwan
- Coordinated squad training and match operations across 30 collegiate athletes and coaching staff.
 - Led team to **4th Place** in the National Inter-University Football Tournament and **5th Place** in the National Futsal Tournament.